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Master of Computer Applications

Semester – III

Data Analytics for Business

Experiment No.: 6

Title: Creating HR analytics dashboards

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Creating HR analytics dashboards

Problem Statement

Organizations today manage large, diverse workforces, making it challenging for HR teams to manually track employee trends and attrition patterns. As an HR Data Analyst in a multinational organization, the goal of this project is to analyze workforce data covering employee demographics, hiring trends, attrition, and workforce distribution, in order to support better workforce planning and employee management decisions.

This project involves developing an interactive HR Analytics Dashboard using Power BI, built on the IBM HR Analytics Employee Attrition & Performance dataset (Kaggle). The dashboard consolidates key metrics — including attrition by department, job role, age group, income, and tenure, along with workforce composition and satisfaction trends — into a single, easy-to-navigate interface, helping HR stakeholders identify why employees leave and make data-driven decisions to improve retention and organizational stability.

Methodology / Steps Followed

1. **Data Collection** – Obtained the IBM HR Analytics Employee Attrition & Performance dataset from Kaggle, containing employee-level records on demographics, job details, compensation, and attrition status.
2. **Data Cleaning & Preparation** – Imported the dataset into Power BI using Power Query, checked data types, removed constant/irrelevant columns (e.g., EmployeeCount, Over18, StandardHours), and verified data quality.
3. **Data Transformation** – Created calculated columns to enable meaningful grouping, such as Age Group, Years at Company Bucket, and Monthly Income Group, to support categorical analysis in visuals.
4. **DAX Measures Creation** – Built key measures including Total Employees, Attrition Count, Active Employees, Attrition Rate, Avg Age, and Avg Years at Company to power the KPI cards and charts.
5. **Dashboard Design** – Designed a two-page interactive dashboard (Overview and Workforce Insights) in Power BI, incorporating KPI cards, donut charts, bar charts, and combo charts to visualize demographics, attrition trends, and workforce distribution.
6. **Interactivity** – Added slicers (Department, Job Role, Education, Gender, Attrition) with cross-page slicer sync and a reset filters button, enabling dynamic filtering and exploration of the data.
7. **Insights & Analysis** – Analyzed the visualized data to extract key findings, such as departments and roles with the highest attrition, and the impact of job satisfaction, work-life balance, and income on employee turnover.

8. **Version Control & Documentation** – Followed the GitHub workflow discussed in class: initialized a repository, structured project files (dataset, .pbix file, screenshots, README), committed changes, and pushed the final project to GitHub for submission.

Dashboard Development

The dashboard was developed in Power BI Desktop as a two-page interactive report, designed to give HR stakeholders a complete view of workforce demographics and attrition at a glance.

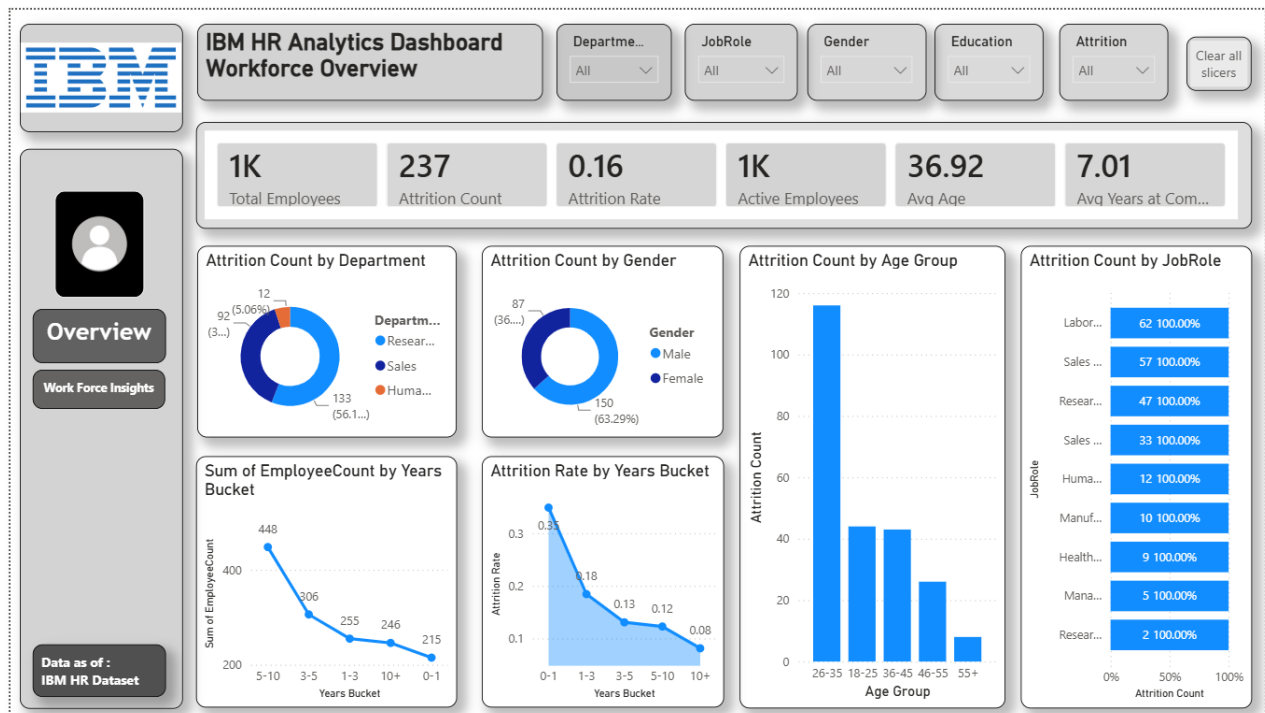
Page 1 – Overview: This page focuses on high-level workforce and attrition metrics. It includes KPI cards for Total Employees, Attrition Count, Attrition Rate, Active Employees, Average Age, and Average Years at Company, along with visuals breaking down attrition by Department, Gender, Age Group, and Job Role. A Hiring Trend line chart and an Attrition Rate by Years at Company chart were added to highlight workforce growth over time and how tenure impacts turnover.

Page 2 – Workforce Insights: This page dives deeper into workforce composition and the drivers behind attrition. It includes visuals for Employees by Education Field, Employees by Job Role (Top 10), Workforce by Department, and Attrition by Marital Status, along with combo charts comparing Monthly Income Group, Job Satisfaction Rating, and Work-Life Balance against Attrition Rate. A Key Insights panel summarizes the major findings drawn from the data.

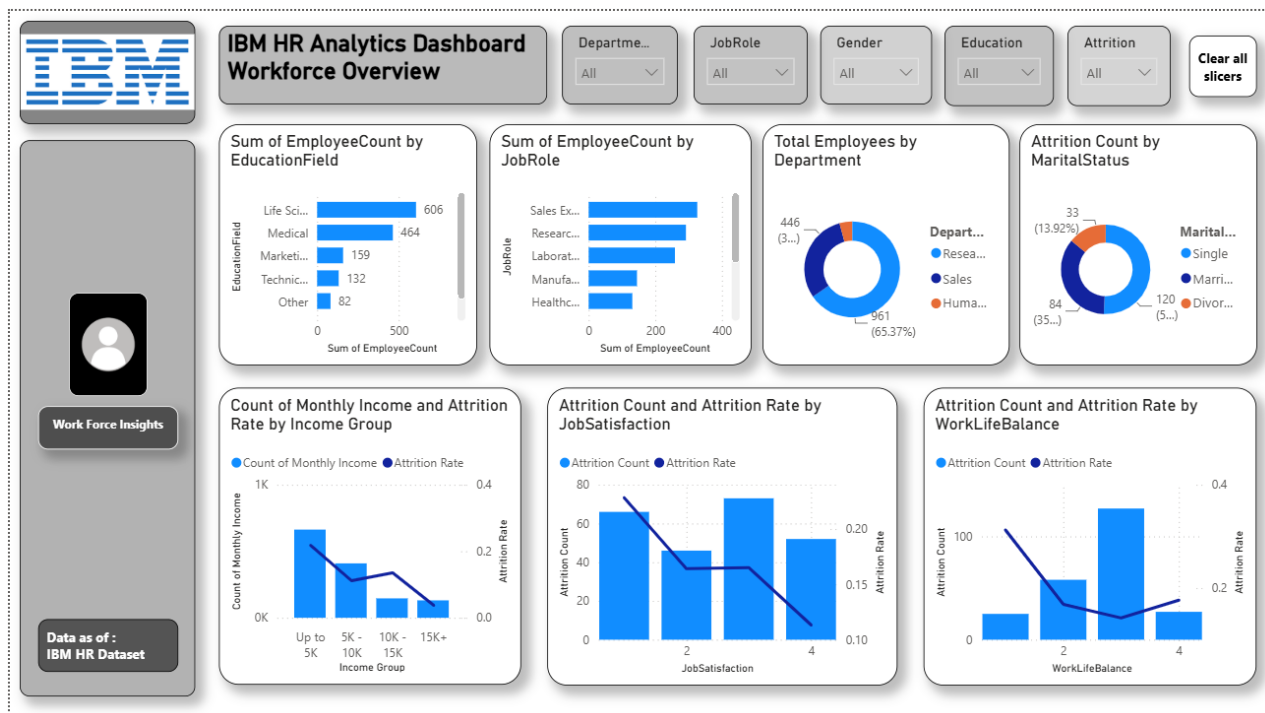
Both pages share a consistent design — an IBM-themed color palette, rounded KPI cards, a common navigation panel, and synced slicers (Department, Job Role, Education, Gender, Attrition) — allowing users to filter both pages simultaneously and explore the data interactively. A Reset Filters button was also added for quick navigation back to the default view.

Dashboard:

Page1:



Page2:



Conclusion

An interactive HR Analytics Dashboard was successfully developed in Power BI using the IBM HR Analytics Employee Attrition & Performance dataset. The dashboard consists of two interactive pages, each offering a distinct perspective on the workforce:

Page 1 – Workforce Overview

- Provides a high-level summary of the organization's workforce and attrition status.
- Displays key KPIs such as Total Employees, Attrition Count, Attrition Rate, Active Employees, Average Age, and Average Years at Company.
- Helps users understand attrition patterns across departments, gender, age groups, and job roles, as well as hiring trends over the years.

Page 2 – Workforce Insights

- Focuses on workforce composition and the underlying drivers of employee attrition.
- Allows HR users to explore employee distribution by education field, job role, department, and marital status.
- Analyzes the relationship between attrition and factors such as monthly income, job satisfaction, and work-life balance, along with a summarized Key Insights panel highlighting major findings.

Interactive slicers (Department, Job Role, Education, Gender, Attrition) synced across both pages, along with a Reset Filters button, make the dashboard user-friendly and allow users to dynamically explore the data. Overall, the dashboard enables HR stakeholders to identify high-risk attrition segments and make informed, data-driven decisions to improve employee retention and support workforce planning.

Learning Outcomes

After completing this experiment, I learned:

- How to create calculated columns for grouping data (e.g., Age Group, Income Group, Years at Company Bucket).
- How to write DAX measures for HR KPI calculations such as Attrition Rate, Active Employees, and Average Tenure.
- How to design and format interactive KPI cards for a dashboard.
- How to create Donut, Bar, Column, Line, and Combo charts to visualize business data.
- How to analyze employee demographics and overall workforce distribution.
- How to identify and analyze key factors influencing employee attrition, such as job satisfaction, income, and work-life balance.
- How to derive and present data-driven HR insights that support workforce planning and retention strategies.

GitHub Repository Link: <https://github.com/deepakMudiraj06/Power-BI/tree/main>

GitHub Project Folder Link: <https://github.com/deepakMudiraj06/Power-BI/blob/main/Assignment%206/Assignment%206.pbix>